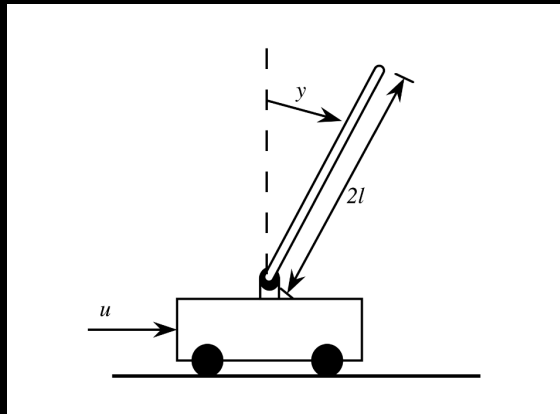


Fuzzy Logic and Control

3

Anders Lyhne Christensen

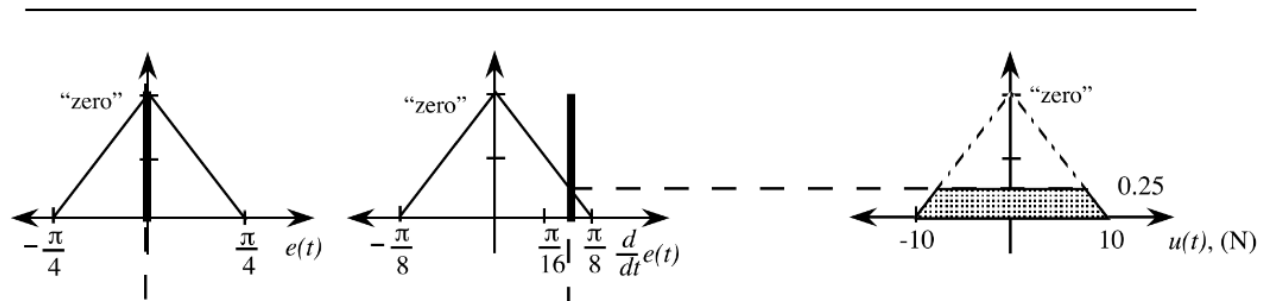
Recap



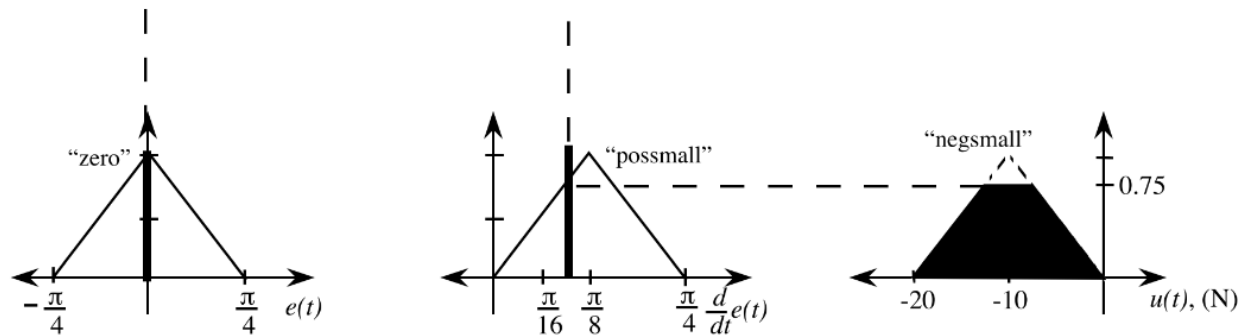
Crisp inputs:

$$e(t) = 0$$

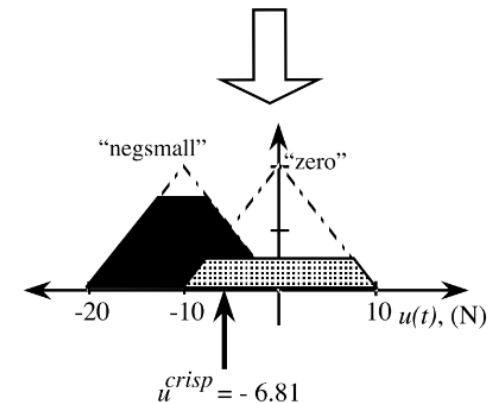
$$\frac{d}{dt}e(t) = \frac{3\pi}{32}$$



If error is zero and change-in-error is zero Then force is zero



If error is zero and change-in-error is possmall Then force is negsmall



Mamdani/standard fuzzy system

If $\tilde{u}_1$ is $\tilde{A}_1^j$ then $\tilde{y}_1$ is $\tilde{B}_1^r$

If $\tilde{u}_1$ is $\tilde{A}_1^j$ and $\tilde{u}_2$ is $\tilde{A}_2^k$ then $\tilde{y}_1$ is $\tilde{B}_1^r$

If $\tilde{u}_1$ is $\tilde{A}_1^j$ or $\tilde{u}_2$ is $\tilde{A}_2^k$ then $\tilde{y}_1$ is $\tilde{B}_1^r$

Mamdani example

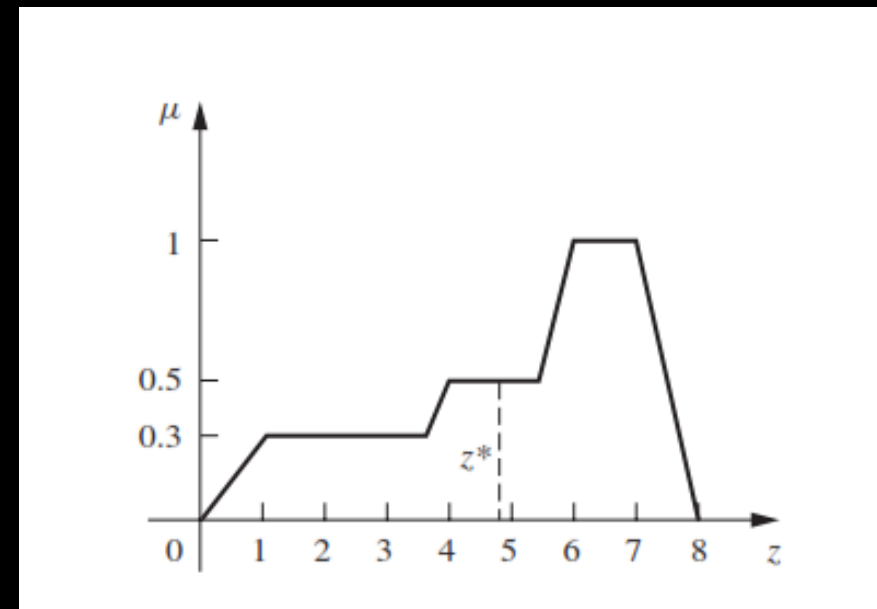
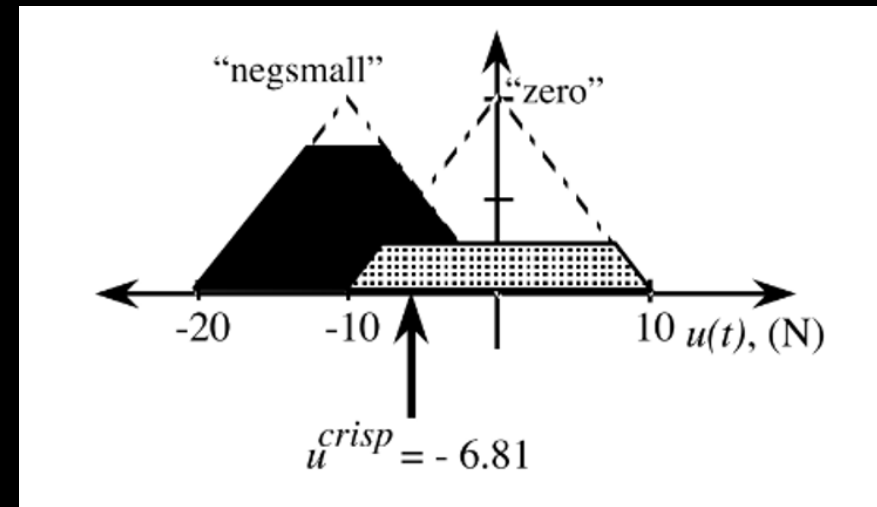
If obs_dist is close then speed is slow

If obs_dir is left and obs_dist is very_close
then direction is sharp_right

If obs_dir is right and obs_dist is very_close
then direction is sharp_left

Mamdani issues

- Mamdani-style inference requires us to find the centroid of a two-dimensional shape by integrating across a continuously varying function. In general, this process is not computationally efficient.
- Michio Sugeno suggested to use a single spike, a singleton, as the membership function of the rule consequent.



Sugeno/Takagi-Sugeno/Functional Fuzzy Systems

- Also known as the TSK fuzzy model (proposed by Takagi, Sugeno, and Kang)

- A typical fuzzy rule in a Sugeno fuzzy model:

If $\tilde{u}_1$ is $\tilde{A}_1^j$ and $\tilde{u}_2$ is $\tilde{A}_2^k$ then $b_i = g(x, y)$

- $\tilde{A}_1^j$ and $\tilde{A}_2^k$: linguistic values/fuzzy sets
- $b_i = g(x, y)$: a **crisp function**
 - usually polynomial in the input variables u_1 and u_2 ($b_i = g(u_1, u_2)$)

Sugeno Fuzzy Systems

if $\tilde{u}_1$ is $\tilde{A}_1^j$ then $b_i = g_i(\cdot)$

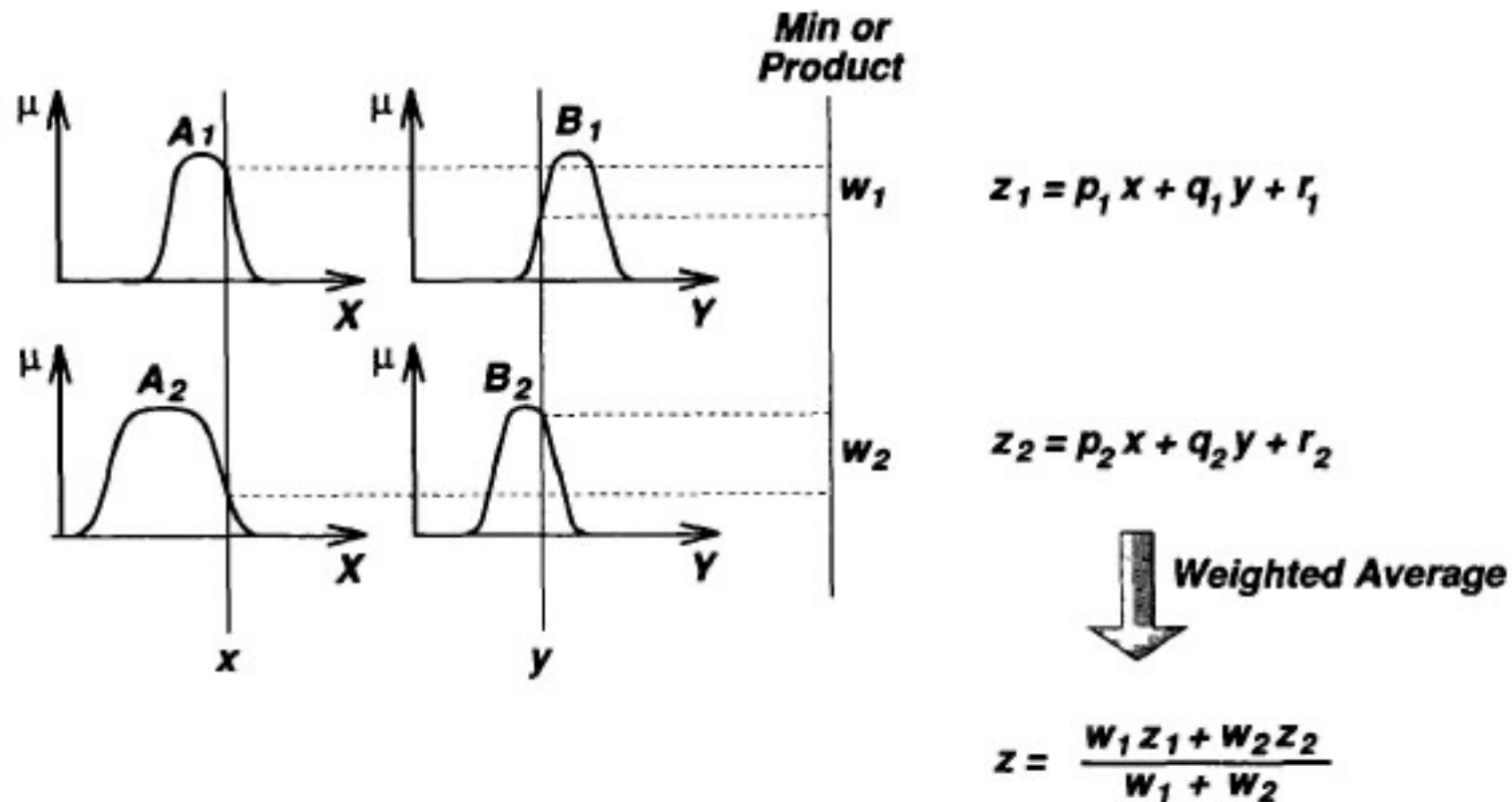
if $\tilde{u}_1$ is $\tilde{A}_1^j$ and $\tilde{u}_2$ is $\tilde{A}_2^k$ then $b_i = g_i(\cdot)$

if $\tilde{u}_1$ is $\tilde{A}_1^j$ or $\tilde{u}_2$ is $\tilde{A}_2^k$ then $b_i = g_i(\cdot)$

Sugeno Example 1/2

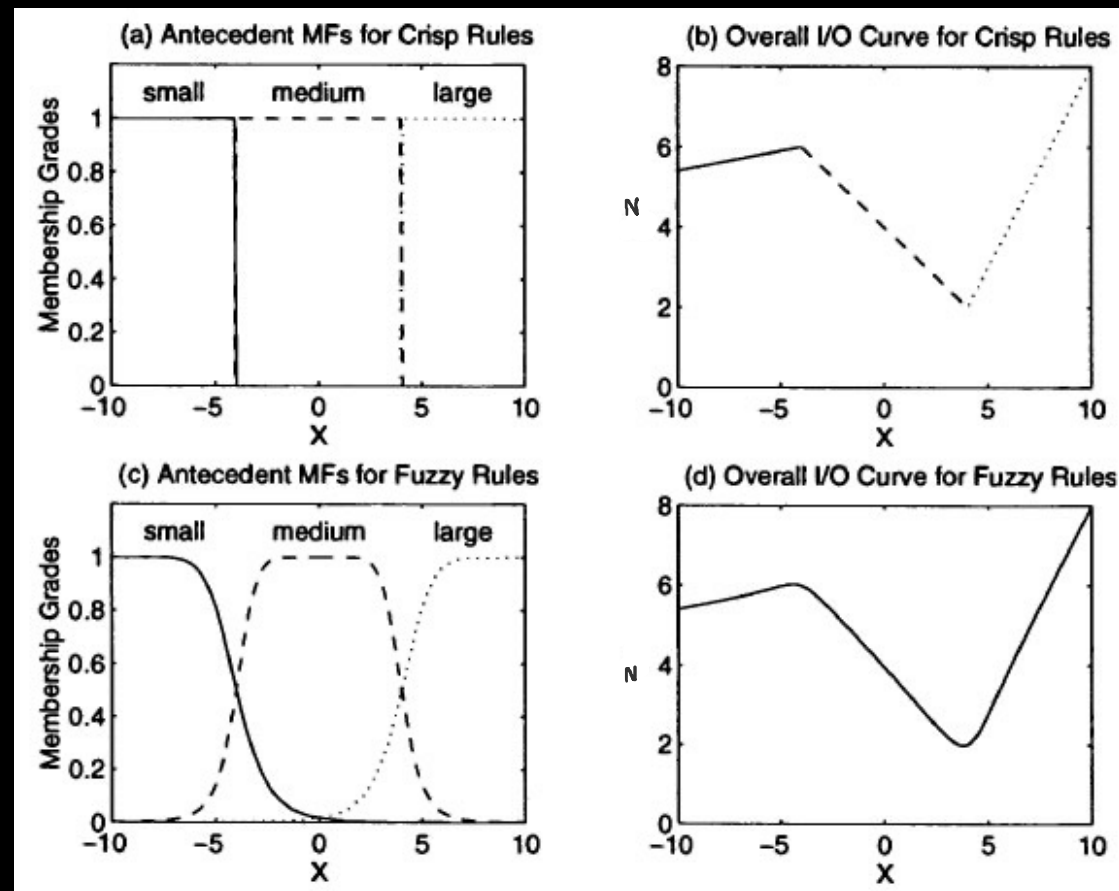
if x is A_1 and y is B_1 then $z = p_1x + q_1y + r_1$

if x is A_2 and y is B_2 then $z = p_2x + q_2y + r_2$



Sugeno Example 2/2

$$\begin{cases} \text{If } X \text{ is small then } Y = 0.1X + 6.4 \\ \text{If } X \text{ is medium then } Y = -0.5X + 4 \\ \text{If } X \text{ is large then } Y = X - 2 \end{cases}$$

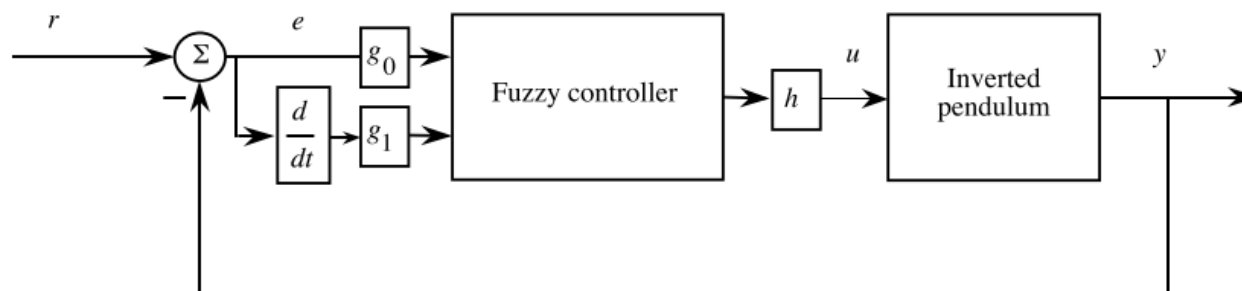
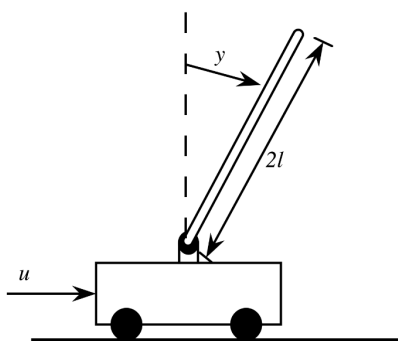


Note that there are yet other types of fuzzy inference systems (FIS), such as:

- Larsen
- Tsukamoto
- Inverse Tsukamoto
- Hybrids

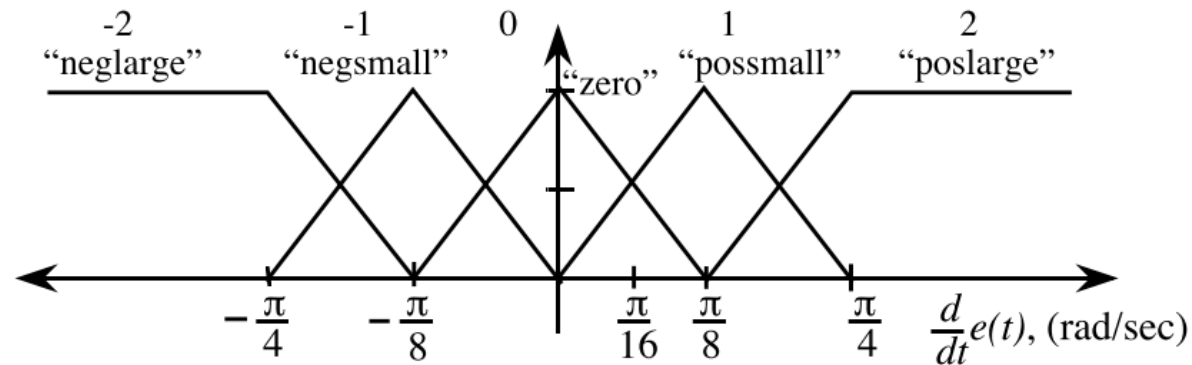
Tuning fuzzy controllers

Input/output gain scaling



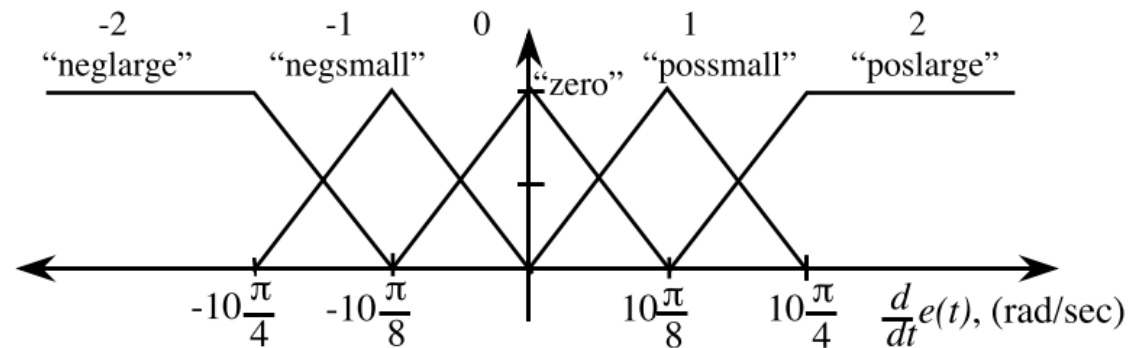
“force” u		“change-in-error” $\dot{e}$				
		-2	-1	0	1	2
“error” e	-2	2	2	2	1	0
	-1	2	2	1	0	-1
	0	2	1	0	-1	-2
	1	1	0	-1	-2	-2
	2	0	-1	-2	-2	-2

Effect of input gain scaling

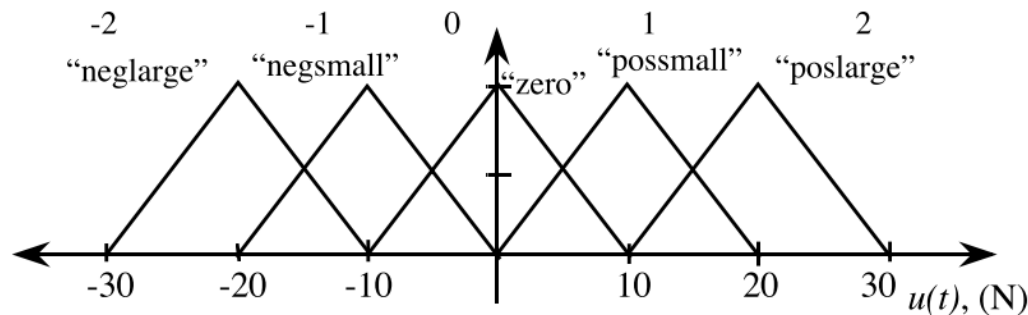


Input membership functions are "spread out" by a factor of $\frac{1}{g_i}$

$g_i = 0.1$

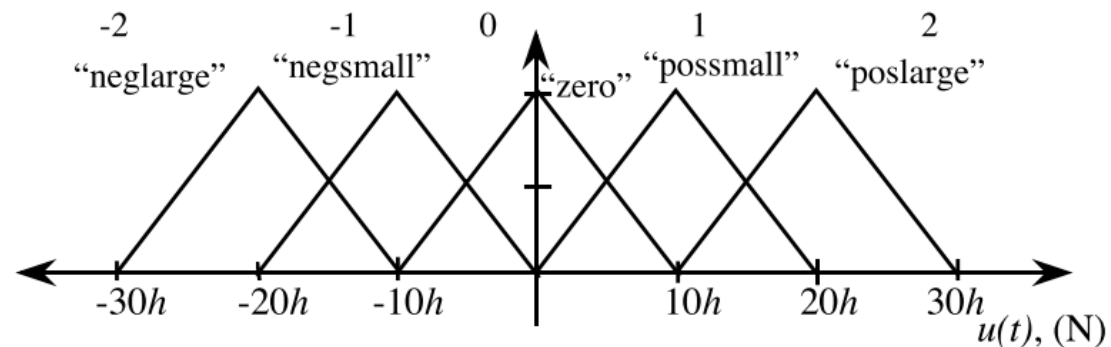


Effect of output gain scaling

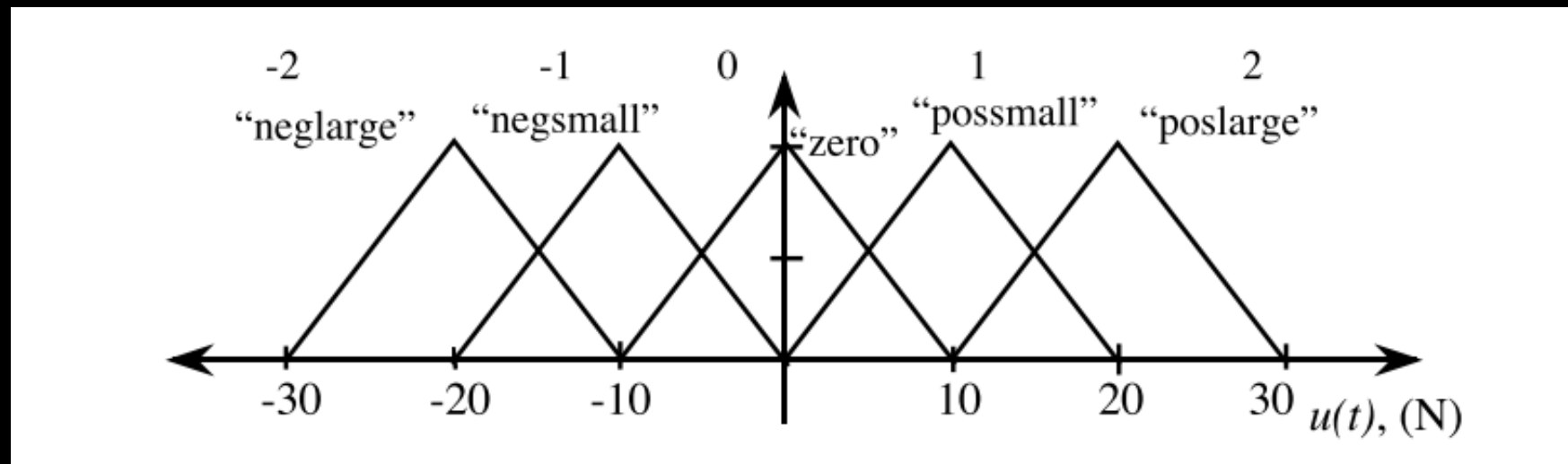


Output membership functions are "spread out" by a factor of h_i

With output scaling



Output membership function tuning



$$c^i = 10i$$

i : linguistic-numerical value.

c^i : center of membership function for i

Output membership function tuning

Base: $c^i = 10i$

With output gain scaling:
 $c^i = 10hi$

Non-uniform scaling:
 $c^i = 5h \operatorname{sign}(i)i^2$



Control surface

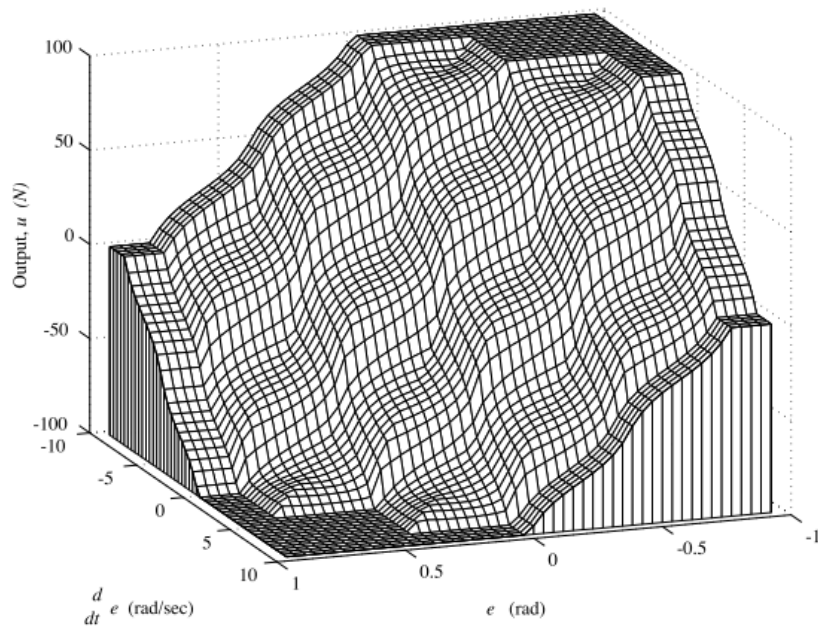


FIGURE 2.34 Control surface of the fuzzy controller for $g_0 = 2.0$, $g_1 = 0.1$, and $h = 5$.

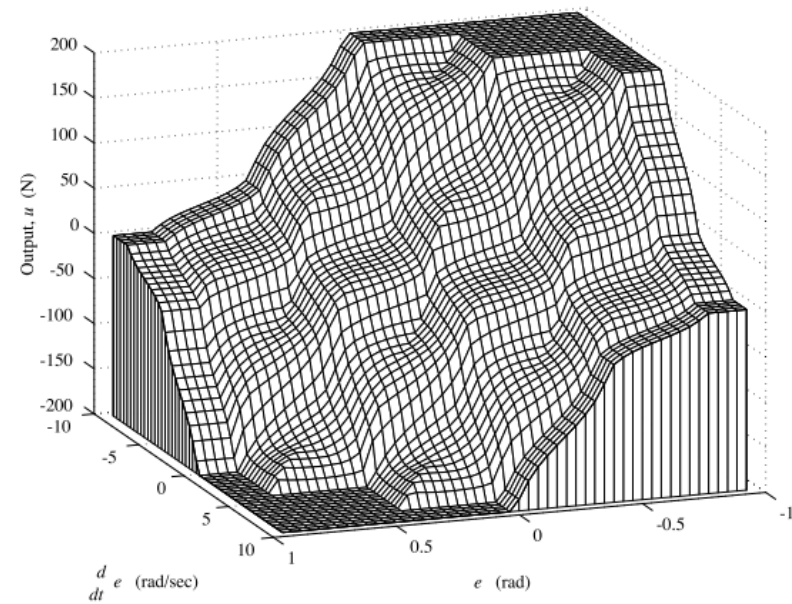


FIGURE 2.35 Control surface of the fuzzy controller for $c^i = 5h \text{sign}(i)i^2$, $h = 10.0$, $g_0 = 2.0$, and $g_1 = 0.1$.

t-norms and s-norms

t-norm, conjunction, "if...*and*..."

If $\tilde{u}_1$ is $\tilde{A}_1^j$ *and* $\tilde{u}_2$ is $\tilde{A}_2^k$ then $\tilde{y}_1$ is $\tilde{B}_1^r$

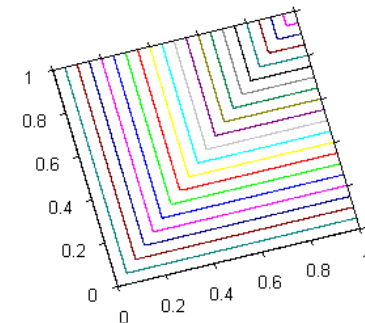
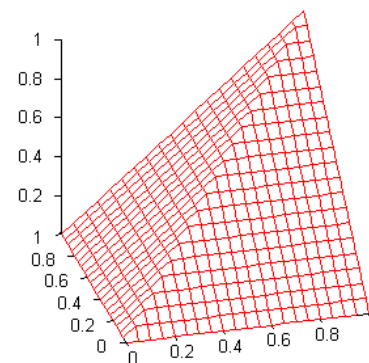
A t-norm is a function $T: [0, 1] \times [0, 1] \rightarrow [0, 1]$ which satisfies the following properties:

- **Commutativity:** $T(a, b) = T(b, a)$
- **Monotonicity:** $T(a, b) \leq T(c, d)$ if $a \leq c$ and $b \leq d$
- **Associativity:** $T(a, T(b, c)) = T(T(a, b), c)$
- The number 1 acts as identity element:

$$T(a, 1) = a$$

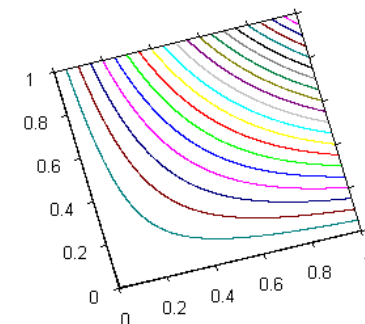
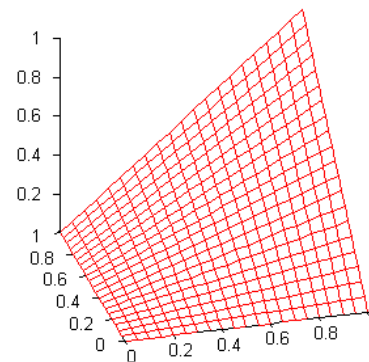
Minimum t-norm:

$$T_{\min}(a, b) = \min(a, b)$$



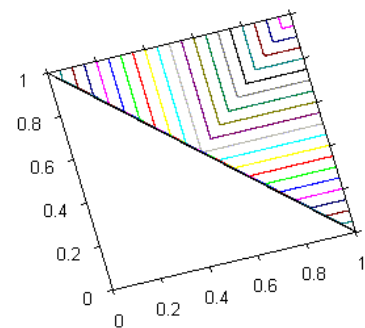
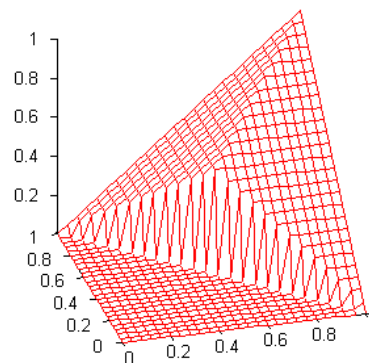
Product t-norm:

$$T_{\text{prod}}(a, b) = a \cdot b$$



Nilpotent t-norm:

$$T_{\text{nM}}(a, b) = \begin{cases} \min(a, b) & \text{if } a + b > 1 \\ 0 & \text{otherwise} \end{cases}$$



s-norms, disjunction, “if...or...”

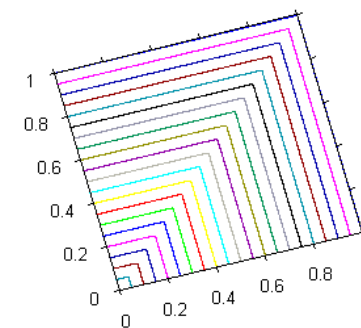
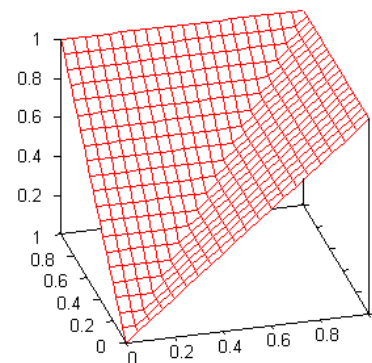
If $\tilde{u}_1$ is $\tilde{A}_1^j$ or $\tilde{u}_2$ is $\tilde{A}_2^k$ then $\tilde{y}_1$ is $\tilde{B}_1^r$

An s-norm (also called a *t-conorm*) must be commutative, monotonous, and associative, but where 0 is the identity element:

$$\perp(a, 0) = a$$

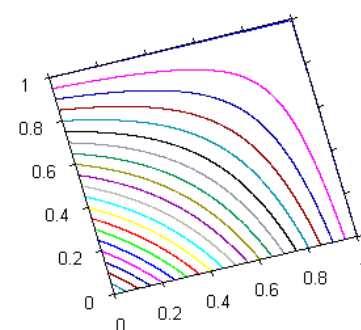
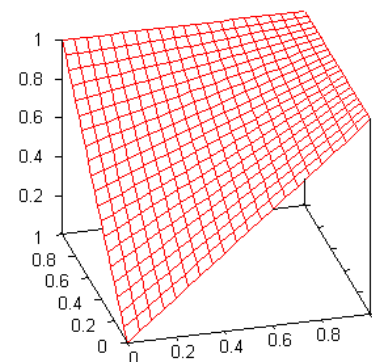
Maximum s-norm:

$$\perp_{\max}(a, b) = \max(a, b)$$



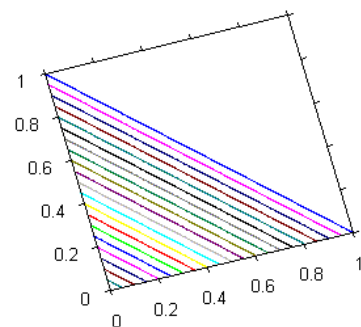
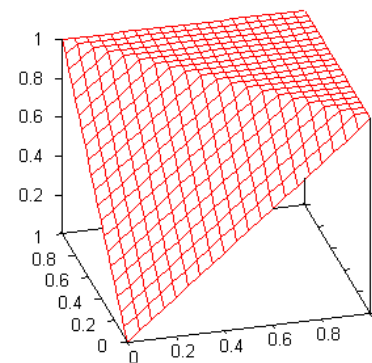
Algebraic/probabilistic sum:

$$\perp_{\text{sum}}(a, b) = a + b - ab$$

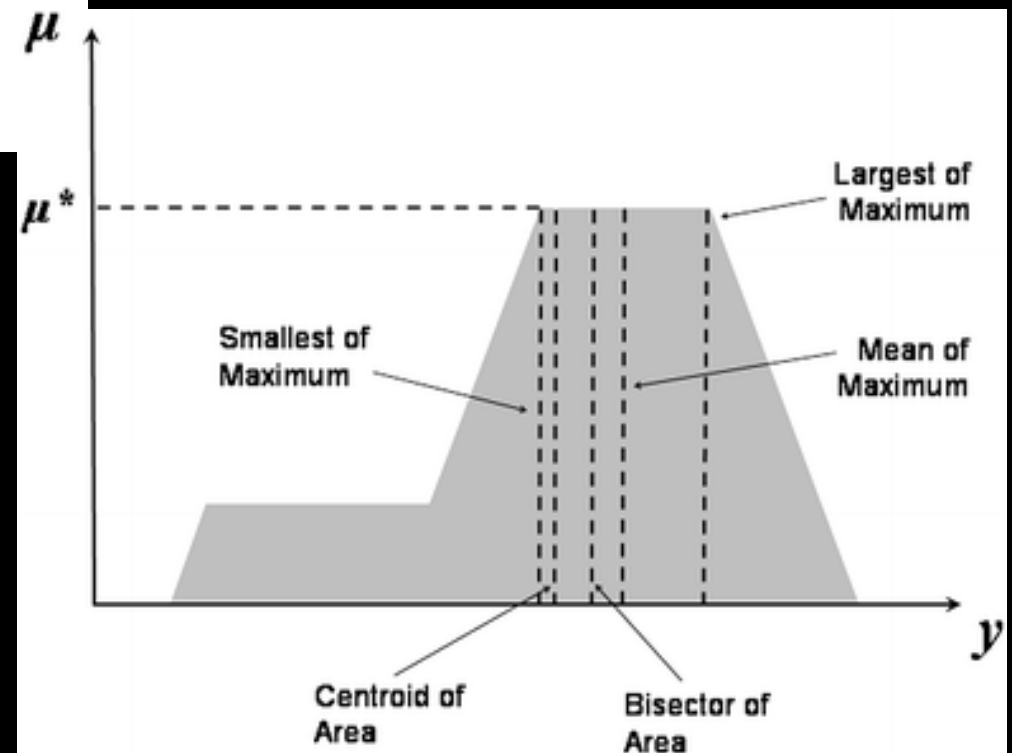
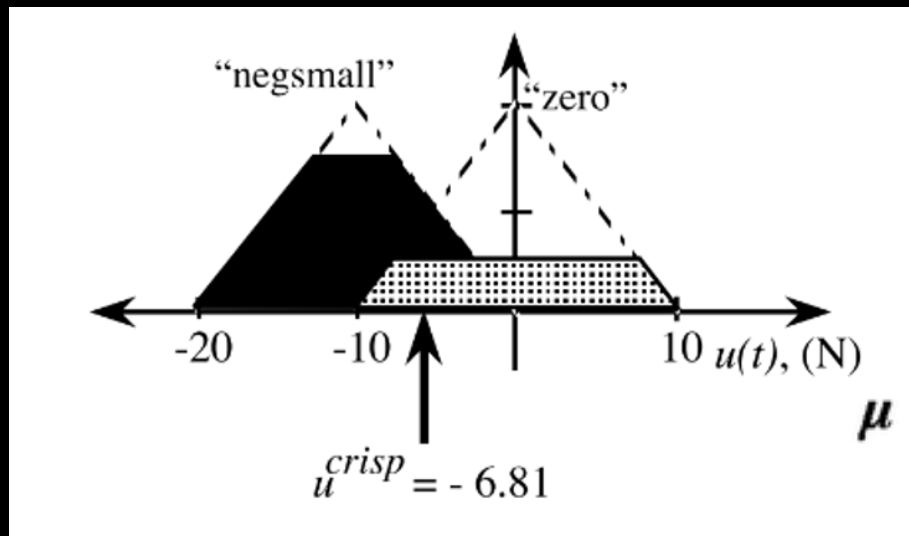


Nilpotent maximum:

$$\perp_{\text{nM}}(a, b) = \begin{cases} \max(a, b) & \text{if } a + b < 1 \\ 1 & \text{otherwise} \end{cases}$$



Defuzzification



Parameters for a fuzzy controller

- Inputs, outputs, universe of discourse
- Membership functions and tuning
- Rule-base
- t-norm and s-norm
- Defuzzification
- Input, output gains

Tuning of fuzzy control

Simple Tuning of Fuzzy Controllers

Eduardo Gómez-Ramírez

Laboratorio de Investigación y Desarrollo de Tecnología A
LIDETEA, Dirección de Posgrado e Investigación, Univers

Design of self tuning fuzzy controllers for nonlinear systems

Rajni Jain ^a✉, N. Sivakumaran ^b✉, T.K. Radhakrishnan ^a✉

Genetic Algorithms for Automated Tuning of Fuzzy Controllers: A Transportation Application

Piero P. Bonissone, Pratap S. Khedkar, Yuto Chen
Information Technology Laboratory

A Robust Self-Tuning Scheme for PI- and PD-Type Fuzzy Controllers

Rajani K. Mudi and Nikhil R. Pal, Member, IEEE

Learning and Tuning Fuzzy Logic Controllers Through Reinforcements

Hamid R. Berenji, Member, IEEE, and Pratap Khedkar

https://link.springer.com/chapter/10.1007/978-3-540-37421-3_7
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<https://ieeexplore.ieee.org/document/746295/#full-text-section>
<https://ieeexplore.ieee.org/abstract/document/159061/>

Basic Design Guidelines

(see Section 2.4.4, page 89 in the book)

and outputs

3. For a variety of reasons, you may choose to try a fuzzy controller (for a discussion of these reasons, see Chapter 1). Be careful to choose the proper inputs to the fuzzy controller. Carefully assess whether you need proportional, integral, and derivative inputs (using standard control engineering ideas). Consider processing plant data into a form that you believe would be most useful for you to control the system if you were actually a “human-in-the-loop.” Specify your best guess at as simple a fuzzy controller as possible (do not add inputs, rules, or membership functions until you know you need them).

For exercises, see [itslearning](#)